

CHAITANYA N.V.K

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Technical Skills

Languages: Java, Python, Go, JavaScript/TypeScript, C++, SQL

Backend & Data: Spring Boot, Node.js, REST, gRPC, Kafka, RabbitMQ, PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS, Kubernetes, Docker, Jenkins, CI/CD, CloudFormation, Prometheus, Grafana

Experience

Workday

Aug 2025 – Present

Software Development Engineer

Atlanta, GA

- **CI/CD API Platform:** Designed and shipped production **/api/v2 REST APIs** for revision sessions, conflict detection, audit reporting, bypass workflows, and push-to-master across **36 PRs**; platform processes approximately **201K API requests** and **1.1M UI/API requests per 30 days**.
- **Security & Reliability:** Implemented **hashed API-key and Bearer authentication** with service/user context separation; hardened push-to-master using client credentials, cancellation, automated bypass handling, and typed **422/503 errors**.
- **Observability & Distributed Tracing:** Added **CloudWatch EMF metrics, dashboards, and alarms** for failures, capacity, orphan detection, and async SLAs; implemented end-to-end trace propagation across **Coffeehouse – CRS – Panda – XORC – GitHub Checks** for cross-service root-cause analysis.
- **Cloud & Async Reliability:** Right-sized production **AWS ECS to 4 vCPU / 16 GB**, authored an **x86-to-Graviton RDS migration** assessment across multiple services, and eliminated indefinitely hung audit workflows using a **10-minute timeout** with step-level async job tracking.

Amazon

Sep 2023 – May 2025

Software Development Engineer

Seattle, WA

- **UNFI Enhancement:** Improved UNFI review and deep dive pages using **AWS Lambda**, streamlining vendor item-store updates and reducing decision-making time from **hours to minutes**.
- **WRAP Backfills:** Led backfill initiatives in the UK, JP, and US, enabling online sales for physical stores and **increasing sales by 20%** in the initial months.
- **WRAP Migration:** Optimized migration workflows and **Lambda configurations**, achieving a **30% reduction in migration time**.
- **Guardrails:** Developed workflow rules and validations, resulting in a **60% decrease in user tickets** by automating error handling and data validation.
- **Risks:** Proactively addressed security and software issues, reducing average turnaround time by **two weeks**.
- **Sourcing Grid:** Designed a new **API and user interface** to streamline configuration, simplifying selection from **five attributes to one**.
- **Grocery Listing:** Enhanced the listing process for the supply chain, cutting user story time from **10 minutes to 4 minutes** for over **25 million global users**.

RE IT Services Inc

Feb 2023 – July 2023

Software Development Engineer

Atlanta, GA

- **Data Pipeline:** Implemented auto archival of database records, resulting in a **90% improvement in query performance**.
- **API Scaling:** Enhanced APIs to support increased load and automated tasks within the **CI/CD pipeline** using **Jenkins** and **Docker**, saving **6 hours of manual work per task**.
- **Data Processing:** Developed **event-driven data processing** solutions using **Kafka** to standardize multi-source data across **18 systems**, eliminating data discrepancies and reducing on-call workload by **11 hours per week**.

Amazon

May 2022 – Aug 2022

SDE Intern at AWS

Metro DC, VA

- **AEA Onboarding:** Collaborated to streamline the **AWS Enterprise Access** onboarding process, reducing average onboarding time by **7 hours per instance**.
- **Topos:** Developed a full-stack console using **Python** to address security issues identified by network scans, enhancing security for **9,000 Amazon services**.

Darwinbox (Cloud-based HRMS platform)

Jun 2018 – Aug 2021

Senior Software Engineer

Hyderabad, AP

- **Microservices Migration:** Led the migration to **microservices architecture** using **Node.js** and **Docker**, reducing throughput time from **6 minutes to 1 minute**, and enabling business workflows as a service.
- **Payslip Customization:** Developed a flexible framework for payslip customization, attracting clients with over **1 million employees** to the platform.
- **Dynamic Reporting Framework:** Implemented a PHP-based reporting engine, handling on-demand data requests for **200,000 employees** in a single operation.
- **MongoDB Optimization:** Executed optimization strategies that saved **\$800K in infrastructure costs** by reducing MongoDB usage.
- **Multi-Payroll Integration:** Created **stateless APIs** for payroll systems, serving **1 million users** across multiple systems.
- **Payroll Automation:** Reduced manual processing hours from **40 to 12** by automating payroll workflows using **Redis** and **RabbitMQ**.
- **Bulk Notifications:** Developed Bash scripts for bulk email and notification processing, supporting communications for over **200,000 users**.
- **Taxation System:** Built a framework for handling multi-country tax systems, attracting **50 new clients** due to enhanced functionality.
- **Recruitment Automation:** Constructed a recruitment automation pipeline using **event emitters**, reducing hiring time by **60%**.

Darwinbox (Cloud-based HRMS platform)

Dec 2017 – Apr 2018

Intern

Hyderabad, AP

- **Chatbot:** Leveraged **Wit.ai** (Facebook NLP tool) to automate tasks for over **1 million customers**, streamlining HR workflows using **Go**.
- **OCR Integration:** Integrated **OCR** into the reimbursement module, automating data extraction from bills and reducing processing time by **90%**.

Education

Georgia State University – Full Scholarship along with GTA

Aug 2021 – Dec 2022

Master of Science in Computer Science; GPA: 3.96/4.0

Atlanta, GA

Vignan University – Merit Scholarship

Aug 2014 – Aug 2018

Bachelor of Technology in Computer Science; GPA: 3.65/4.0

Guntur, AP, India

Personal Projects

Gumo | Node.js, MongoDB, NPM

Apr 2021

- Built a web crawler and scraper to extract data from nested dynamic web pages and support downstream knowledge-mining applications.

NFL Health & Safety | Python, Pandas, NumPy

Dec 2021

- Developed helmet segmentation, player tracking, and labeling methods for the NFL-AWS player safety challenge; findings later published on arXiv.

Volunteering

Mentors Without Borders

Feb 2026 – Present

Volunteer Instructor

Remote

- Mentor junior high school students in Ghana through weekly remote sessions, teaching **web development and programming fundamentals** with hands-on exercises and structured projects.

Research Publications

- **Nuthalapati, V.K.C.** *Balancing Microservices and Monolithic Architectures*. IEEE Computer Society Santa Clara Valley Chapter, *Feedforward*, Vol. 5, No. 3, pp. 28–33, Jul 2026. [arXiv:2607.03898](#)
- **Mathur, M., Chandrashekhar, A.B., Nuthalapati, V.K.C.** *Real Time Multi-Object Detection for Helmet Safety*. arXiv, May 2022. [arXiv:2205.09878](#)
- **Agrawal, A., Bhyravarapu, S., Nuthalapati, V.K.C., et al.** *Matrix Hashing with Two Level of Collision Resolution*. 2018 8th International Conference on Cloud Computing, Data Science & Engineering (Confluence), IEEE, 2018. DOI: [10.1109/CONFLUENCE.2018.8442607](#)
- **Vattikuti, B., Nuthalapati, N.V.K.C., Jukuntla, A.** *Installation Automation Tool*. Advances in Intelligent Systems and Computing, Springer, 2018. DOI: [10.1007/978-981-10-7868-2_24](#)

 [Google Scholar Profile](#)